



Shivansh Malik

Computer Science Undergraduate · Cyber Security & Applied AI
Building intelligent, security-minded systems end to end.

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PROFILE

Cyber Security-focused Computer Science undergraduate (CGPA 8.42/10) who designs and ships intelligent, security-minded systems end to end — from ML-driven intrusion detection to predictive healthcare, database-backed platforms, and applied Generative AI. Strong in data structures & algorithms, computer networks, operating systems, and database systems, with hands-on coursework spanning artificial intelligence, data analytics, and discrete mathematics. Curiosity spans probability, quantitative finance, and quantum computing. A national-level athlete and club lead who brings discipline, clear communication, and initiative to every team.

EDUCATION

YEAR	DEGREE / CERTIFICATE	INSTITUTE	CGPA/%
2024–Present	B.Tech, Computer Science & Engineering — Cyber Security	Manipal Institute of Technology, MAHE, Bengaluru	8.42/10
2023	CBSE (XII), Senior Secondary	Woodrow Senior Secondary School, Bareilly	83%
2021	CBSE (X), Secondary	Kendriya Vidyalaya, IFFCO — Aonla, Bareilly	94%

Semester I	Semester II	Semester III	Semester IV
8.85	8.41	7.48	8.95

Academic Distinctions: A+ in Design & Analysis of Algorithms, Introduction to Artificial Intelligence, Object-Oriented Software Development Lab, Data Visualisation, Environmental Studies, Fundamentals of Electrical Engineering, and Introduction to Object-Oriented Programming Lab; A in Database Systems Lab, Operating Systems Lab, Digital Systems Lab, Programming for Problem Solving Lab, Engineering Mechanics & Smart Buildings, Applied Chemistry for Engineers, Applied Physics for Engineers, Workshop Practice, and Communication Skills in English.

Relevant Coursework:

Data Structures & Lab	Design & Analysis of Algorithms	Operating Systems & Lab
Database Systems & Lab	Object-Oriented Programming & Lab	Digital Systems & Computer Organization
Data Communication & Computer Networks	Discrete Mathematical Structures	Computational Mathematics I & II
Probability & Optimization	Introduction to Artificial Intelligence	Data Analytics & Visualisation

Additional Coursework (First-Year Engineering Foundation):

Applied Physics for Engineers	Applied Chemistry for Engineers	Fundamentals of Electrical Engineering
Fundamentals of Electronics	Basic Mechanical Engineering Science	Engineering Mechanics & Smart Buildings
Workshop Practice	Computer-Aided Engineering Graphics	Communication Skills in English
Environmental Studies	Universal Human Values & Professional Ethics	Human Rights & the Constitution

TECHNICAL SKILLS

Languages	C, C++, Java, Python, JavaScript, HTML/CSS, SQL
Gen AI & LLM Engineering	Prompt engineering, RAG, embeddings, LangChain (LCEL, memory, tools), LangGraph (state machines, conditional edges, agentic workflows), Groq/Gemini/Ollama APIs, ChromaDB, Streamlit, Pydantic
Machine Learning	scikit-learn, XGBoost, Pandas, NumPy, Random Forest, SMOTE, feature engineering, model evaluation
Data Analytics & Visualization	Exploratory data analysis, Matplotlib, Seaborn, statistical analysis
Databases	MySQL, Oracle SQL, MongoDB — schema design, transactions & concurrency control, query optimization
Computer Networks	OSI & TCP/IP models, IPv6 addressing & subnetting, routing fundamentals, socket programming
Systems & OS	Linux internals, POSIX threads & IPC, process synchronization, shell scripting
Cyber Security	Network traffic analysis (Wireshark), vulnerability & threat awareness, secure system design, ML-based intrusion detection
DSA & Competitive Programming	Working through a structured DSA sheet in C++ (arrays, trees, graphs, DP, greedy) with CLRS as reference; active on LeetCode and Codeforces, targeting 300+ problems solved and a 1400+ CF rating
Probability, Optimization & Quant Math	Probability theory, statistical optimization, curiosity in quantitative finance and algorithmic trading concepts
Core CS Fundamentals	DSA, OOP, DBMS, OS, Computer Networks, AI/ML, Discrete Mathematics, Digital Systems & Computer Organization
Tools & Platforms	Git/GitHub, JavaFX, Wireshark, VS Code, Linux/Unix CLI
Strengths	Leadership, communication, teamwork, public speaking, mentoring

EXPERIENCE

Generative AI Intern — Softpro School of AI

SSAI-101 · 46-Day AI Development Summer Training

Jun 2026 – Jul 2026
Onsite · Lucknow, Uttar Pradesh

OBJECTIVE	Build practical, production-style Generative AI and LLM engineering skills across a structured curriculum, progressing from prompt engineering through retrieval-augmented generation to agentic, tool-using workflows.
APPROACH	- Built and deployed multi-provider LLM applications on the Groq API (with Gemini/Ollama fallbacks), applying zero-shot, few-shot, and chain-of-thought prompting with Pydantic-validated structured outputs. - Implemented a semantic search and RAG pipeline from scratch using sentence-transformer embeddings and a Chroma vector database. - Progressing into agentic AI with LangChain (LCEL chains, prompt templates, memory) and LangGraph (state machines, conditional edges, tool-calling agents) to orchestrate multi-step LLM workflows.
RESULT	Shipped two Streamlit mini-projects — an AI content summarizer (PDF/URL input) and a citation-aware "Chat With Your Documents" RAG assistant — and is now building an agentic Research Assistant with LangGraph (see Projects).

Stack: Python, LangChain, LangGraph, Groq/Gemini/Ollama APIs, Sentence-Transformers, ChromaDB, Streamlit, Pydantic

PROJECTS

Research Assistant Agent — LangGraph Agentic Research Tool

Softpro AI Internship · Major/Capstone Project

Jul 2026 – Present

OBJECTIVE	Build a LangGraph agent that takes a research topic and autonomously searches, reads, and synthesizes sources into a cited mini-report, with its reasoning visible in the UI.
APPROACH	- Designing a LangGraph state machine with conditional edges to orchestrate a Think → Act → Observe loop across tool calls (search, retrieval, summarization). - Implementing <code>@tool</code>-decorated retrieval/search functions bound to the LLM via <code>bind_tools</code>, with structured, citation-aware output. - Building a Streamlit UI that streams the agent's intermediate reasoning and tool calls live alongside the final cited report.
RESULT	In active development — targeting a working end-to-end demo with live agent-trace visualization.

Stack: Python, LangChain, LangGraph, Groq API, Streamlit, Pydantic

OBJECTIVE	Build a Streamlit app that turns a PDF, article URL, or transcript into a structured summary (TL;DR, key points, action items, tags).
APPROACH	- Extracted text from PDFs (pdfplumber) and scraped article URLs (requests + BeautifulSoup) behind a single input flow. - Used Pydantic-validated structured output from the Groq API so the summary always parses into consistent fields, not free-form text.
RESULT	Shipped a working Streamlit app accepting either input type and returning a structured, consistently-formatted summary.

Stack: Python, Groq API, Pydantic, pdfplumber, BeautifulSoup, Streamlit

OBJECTIVE	Build a Streamlit app that lets users upload PDFs/text documents and get answers grounded strictly in the uploaded content.
APPROACH	- Chunked and embedded documents with sentence-transformer models, persisting vectors in a local ChromaDB store. - Built a RAG pipeline over the Groq API with a "Sources used" citation panel on every answer.
RESULT	Delivered a working document Q&A assistant with multi-file support and a chat-history sidebar.

Stack: Python, LangChain, ChromaDB, Sentence-Transformers, Groq API, Streamlit

OBJECTIVE	Build a fully automated customer feedback system for a local grocery mart, from form submission to owner notification and reporting.
APPROACH	- Built the backend end-to-end in Google Apps Script bound to a Google Sheet, managed via the clasp CLI, with an auto-generated Google Form. - Implemented instant email alerts on every submission and a weekly digest (time-driven trigger) with an .xlsx export of all responses.
RESULT	Deployed a live, self-sustaining system, verified end to end; QR-code posters (A3/A4) drive customers to the form in-store.

Stack: Google Apps Script, Google Sheets/Forms API, clasp, JavaScript

OBJECTIVE	Build a modular, end-to-end ML pipeline flagging malicious network traffic on the NSL-KDD benchmark (41 features; binary & 5-class — Normal/DoS/Probe/R2L/U2R).
APPROACH	- Countered severe class imbalance with SMOTE; compared correlation, ANOVA-SelectKBest, and Random-Forest feature selection. - Benchmarked six classifiers to identify the strongest architecture for deployment. - Documented the preprocessing and evaluation methodology for reproducibility across binary and multi-class settings.
RESULT	Auto-selected and serialized the strongest model (~78% macro-F1) for deployment.

Stack: Python, scikit-learn, XGBoost, imbalanced-learn, Pandas

OBJECTIVE	Rebuilt as a Random-Forest symptom-to-condition engine behind a FastAPI ML microservice, with a JavaFX client for free-text intake, explainable predictions, and hospital lookup.
APPROACH	- Added an NLP intake path: a sentence-transformer embedding matcher links free-text symptom descriptions to canonical symptoms, so users aren't limited to a checkbox list. - Made every prediction explainable with per-request SHAP (TreeExplainer) breakdowns of which reported symptoms actually drove that diagnosis, not just a bare label. - Built a specialty-aware hospital locator (haversine distance + disease→specialty mapping) so recommendations route to the relevant department, not just the nearest building.
RESULT	Shipped a working FastAPI backend + JavaFX desktop client end to end; ~97.6% held-out test accuracy (41 conditions, 132 symptoms).

Stack: Java, JavaFX, Python, FastAPI, scikit-learn, SHAP, Sentence-Transformers

OBJECTIVE	Rebuilt around two distinct problems: seat-level double-booking under concurrency, and database overload from a thundering herd during high-demand ticket drops.
APPROACH	- Used PostgreSQL row-level locking (SELECT ... FOR UPDATE SKIP LOCKED for holds, FOR UPDATE for confirms) plus a UNIQUE (seat_id) constraint and idempotency keys as defense-in-depth against double-booking. - Built a Redis-backed virtual waiting room (sorted-set FIFO queue + TTL-based admission) that rate-limits access to the booking flow for high-demand events, the pattern real ticketing platforms use for ticket drops. - Wrote an async FastAPI backend (asyncpg) with a WebSocket-driven live seat map, and a concurrency stress-test harness that fires real concurrent requests at one seat to empirically verify correctness.
RESULT	200 simulated concurrent users through the full queue+booking flow on one seat → exactly 1 confirmed booking, verified at the database level.

Stack: PostgreSQL, Python, FastAPI, asyncpg, Redis, WebSockets

POSITIONS OF RESPONSIBILITY

- Lead educational outreach for a flagship technical club — design and teach quantum-computing sessions for juniors, and help run events and workshops.
- Curate learning material bridging classical CS foundations with quantum computing concepts for a general undergraduate audience.
- Coordinate with faculty advisors to align club activities and session content with departmental academic goals.

- Captain the institute squad — leading training, selection, and match strategy — to multiple tournament wins.
- Balance competitive-sport discipline with a rigorous academic schedule, translating athletic focus into consistent performance.
- Mentor junior players joining the institute team, focusing on technique, match temperament, and tournament readiness.

- Sharpen data structures & algorithms with regular problem-solving; compete in hackathons and competitive-programming contests.
- Collaborate with peers on club-organized coding challenges and knowledge-sharing sessions.
- Contribute to peer code reviews and pair-programming sessions to reinforce good software practices.

- Represented the student body as the elected head, acting as the primary liaison between students and faculty/administration.
- Led discipline and conduct during assemblies, morning prayers, and school-wide events, setting the standard for student behavior.
- Coordinated inter-house competitions and annual day/cultural functions alongside teachers and fellow student leaders.

CERTIFICATIONS

CERTIFICATION	ISSUER	YEAR
Linux on IBM LinuxONE	IBM SkillsBuild	2026
Integrating with Structured Databases	IBM SkillsBuild	2026
Data Fundamentals	IBM SkillsBuild	2025
Fundamentals of Java Programming	Board Infinity / Coursera	2025

ACHIEVEMENTS & EXTRA-CURRICULAR

Academic	- Class X Merit Scholarship from IFFCO for academic excellence. - NTSE Stage I — 150/200 (cleared cut-off); strong KVPY score. 13 A+/A grades across first-year foundation and core CS coursework (full list under Education — Academic Distinctions).
Sports	2× National-level Table Tennis player; captained MIT's team to multiple tournament titles.
Cultural	Professional Bharatanatyam dancer — represented Aonla twice, with a scholarship awarded.

INTERESTS

AI/ML · Cyber Security · Quantum Computing · Quantitative Finance · Competitive Programming · Chess · Reading History